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Uganda Certificate of Education
END OF TERM 1 ASSESSMENT 2026

S.3 PHYSICS

Paper 1

1 ½ hour

INSTRUCTIONS TO CANDIDATES:

This paper consists of three items;

Respond to only two items.

Item 1.

A man set off from Arua to a town, a destination 250 km away from his home, to be at his workplace at 8:00 am. The speed limit on the road is 80 km/h. He set off on the journey without fastening his seatbelt and accelerated uniformly to a velocity of 72 kmh^{-1} and maintained this velocity for the first 2 hours before reaching Murchison Falls National Park. In the park, an elephant appeared from nowhere and started crossing the road. He suddenly applied the brakes, decelerating uniformly at 2 ms^{-2} . The sudden deceleration made him jerk forward, almost crashing into his windscreen. The man stopped for 45 minutes and resumed his journey, reaching his workplace on time.

Task:

- (a) Determine if the man exceeded the recommended speed limit before stopping.
- (b) What distance had he covered when he reached the national park.
- (c) Explain what made the man jerk forward.
- (d) Explain the importance of fastening a seat belt before driving a car.

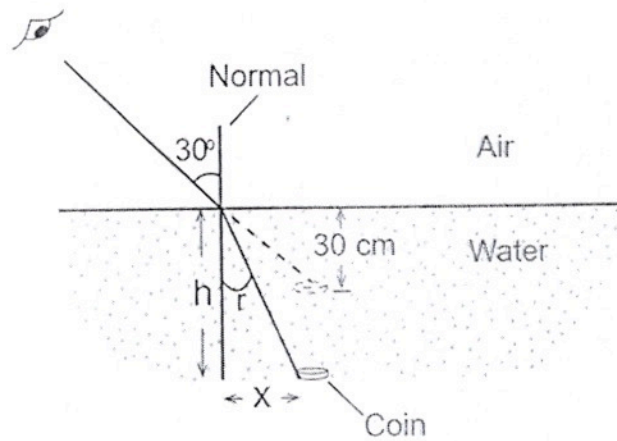
Item 2.

During a football match, a player kicked a ball, which moved forward but was rolling backwards. The striker (S) of his team rushed to receive the ball and collided with the defender (D) of the opponent team. The two men collided and moved together for some time before falling. Sustaining some injuries as they fell on the grass in the pitch. They both missed the ball, and it went offside.

Hint:

Analysis of the video footage showed that:

Support:



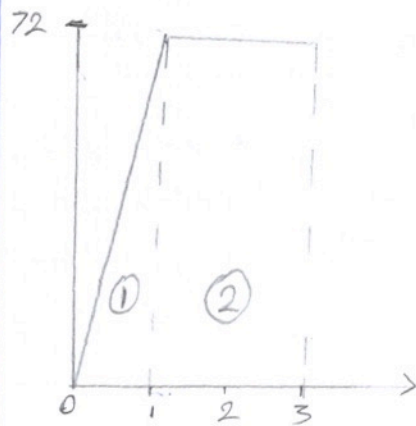
Task:

As a physicist:

- Explain why the hand of the girl appeared bent in the water.
- Determine the real depth, h , of the pond.
- If a ray of light is incident at an angle of 30° to the normal, as shown in the diagram, determine the distance, x , of the coin from the normal.
- State any two other observations in real life of refraction of light.

Item 1

a)



$$\begin{aligned}
 \text{Distance} &= \text{Area 1} + \text{Area 2} \\
 &= \left(\frac{1}{2} b \times h\right) + (L \times W) \\
 &= \left(\frac{1}{2} \times 1 \times 72\right) + (72 \times 2) \\
 &= \left(\frac{1}{2} \times 72\right) + 144 \\
 &= 36 + 144 \\
 &= 180 \text{ km}
 \end{aligned}$$

$$\begin{aligned}
 \text{Speed} &= \frac{D}{T} \\
 &= \frac{180}{3} \\
 &= 60 \text{ km/hr}
 \end{aligned}$$

$\therefore$ The man did not exceed the recommended speed limit before stopping.

b) Distance covered when he reached the national park

$$\begin{aligned}
 &= \text{Area 1} + \text{Area 2} \\
 &= \left(\frac{1}{2} b h\right) + (L \times W) \\
 &= \left(\frac{1}{2} \times 1 \times 72\right) + (72 \times 2) \\
 &= \frac{72}{2} + 144 \\
 &= 36 + 144 \\
 &= 180 \text{ km}
 \end{aligned}$$

$\therefore$ He had covered a distance of 180 km when he reached the national park.

Item 2

a) The ball rolled in the opposite direction as it moved the air due to the frictional force in the air when it has to overcome as it moves in the air. This is the same reason why the bodies of birds are stream lined.

b) Data $S = 60\text{kg}$, $V = 5\text{ms}^{-1}$, $D = 80\text{kg}$, $V = 10\text{ms}^{-1}$

$$M_S U_S + M_D U_D = (M_S + M_D) V$$

$$60 \times 5 + 80 \times 10 = (60 + 80) V$$

$$300 + 800 = (140) V$$

$$1100 = 140 V$$

$$\frac{1100}{140} = \frac{1100}{140}$$

$$V = 7.857\text{ms}^{-1} / 7.86\text{ms}^{-1}$$

$\therefore$ Their common velocity after the collision is 7.86

c) Loss in kinetic energy

$$= \frac{1}{2} m_D v^2 - \frac{1}{2} m_S v^2$$

$$= \frac{1}{2} 80 \times 7.86^2 - \frac{1}{2} 60 \times 7.86^2$$

$$= 40 \times 7.86^2 - 30 \times 7.86^2$$

$$= 40 \times (7.86 \times 7.86) - 30 \times (7.86 \times 7.86)$$

$$= 40 \times 61.78 - 30 \times 61.78$$

$$= 2471.2 - 1853.4$$

$$= 617.8\text{J} / 618\text{J}$$

$\therefore$ The loss in kinetic energy is 618J

d) Football players are advised to wear protective gear (shin guards) around their legs during football matches because the shin guards absorb the impulsive force applied on their legs hence reducing the impulse received on their legs. This reduces the injuries / pain received on their legs.